



UNDERSTANDING USED CAR PRICES

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Introduction :

Buying a used car has become a popular choice because it offers more affordable options than buying a new car. However, used cars differ in many aspects, such as manufacturing year, mileage, condition, manufacturer, and vehicle type, which can affect their prices. In this project, a dataset of 426,880 used car listings from the United States will be analyzed to better understand the factors that influence used car prices.



Problem Statement :

With 426,880 used car listings across the United States, buyers have many different options with different prices and specifications. This makes it difficult to understand what affects used car prices and to compare similar cars to determine a fair market price. As a result, buyers may make poor purchasing decisions.



BMW 2001 Price Comparison

	manufacturer	model	year	price	odometer	condition	transmission	type	size
290389	bmw	530i m5	2001	550	190000.0	fair	automatic	sedan	mid-size
364025	bmw	3 series	2001	15500	95770.0	excellent	manual	convertible	mid-size

Jeep 2016 Price Comparison

	manufacturer	model	year	price	odometer	condition	transmission	type	size
331947	jeep	patriot	2016	500	154119.0	excellent	automatic	sedan	mid-size
30486	jeep	wrangler unlimited	2016	65000	42600.0	like new	automatic	SUV	full-size

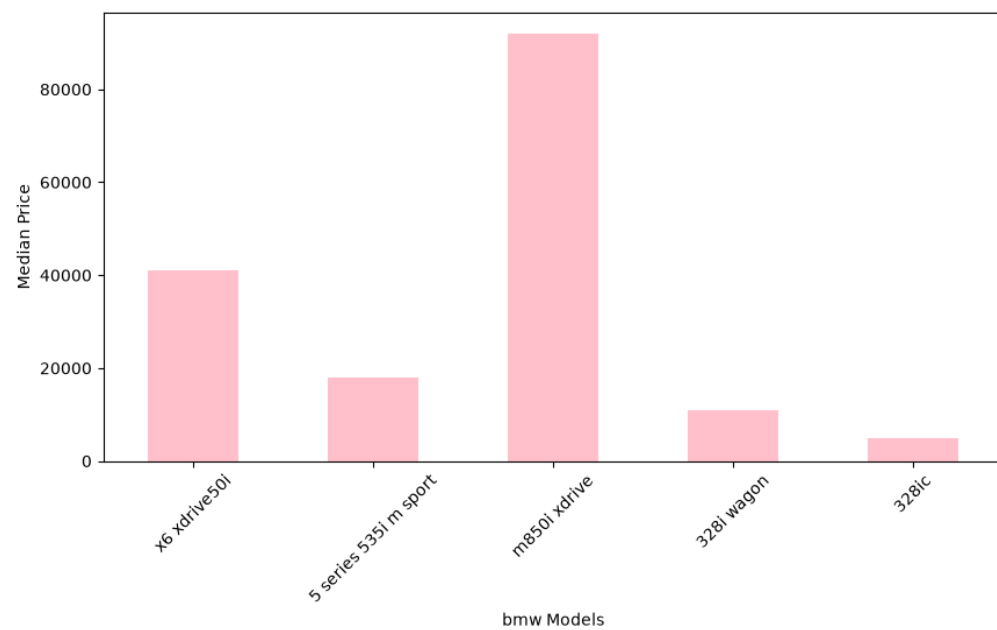
Key Finding :

Even cars from the same manufacturer and year can have different prices because of the model.

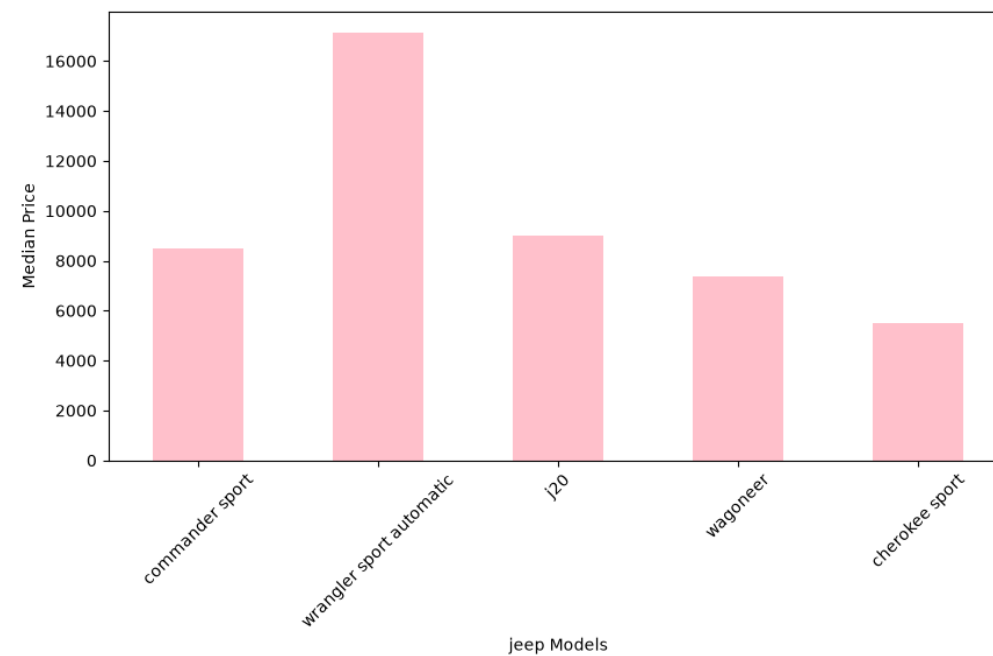
OBJECTIVE 1
HOW CAN TWO CARS FROM THE SAME MANUFACTURER AND YEAR HAVE DIFFERENT PRICES?



Model affect on BMW 2001 Price :



Model affect on Jeep 2016 Price :



Honda civic 2015 Price Comparison

	manufacturer	model	year	price	odometer	condition	transmission	type	size
157889	honda	civic sedan	2015	8450	100000.0	excellent	automatic	sedan	compact
163859	honda	civic sedan	2015	14995	39623.0	good	automatic	sedan	compact

Toyota Camry 2005 Price Comparison

:	manufacturer	model	year	price	odometer	condition	transmission	type	size
380241	toyota	camry	2005	999	264000.0	good	automatic	sedan	mid-size
66570	toyota	camry	2005	7999	123240.0	excellent	automatic	sedan	mid-size

Key Finding :

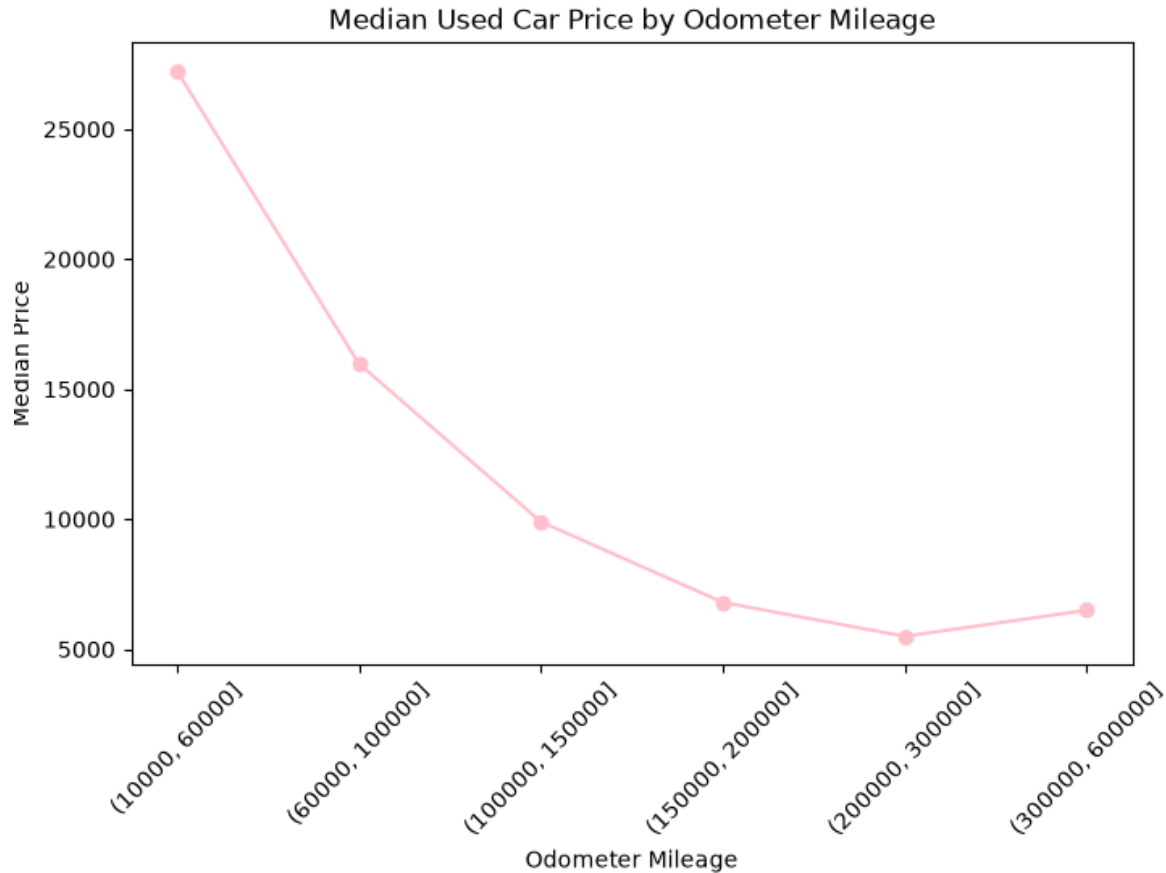
Even cars from the same manufacturer, model, and year can have different prices because of factors such as mileage, condition, transmission, and vehicle type.

OBJECTIVE 2
HOW CAN TWO
CARS FROM THE
SAME
MANUFACTURER,
MODEL, AND
YEAR HAVE
DIFFERENT
PRICES?



OBJECTIVE 3

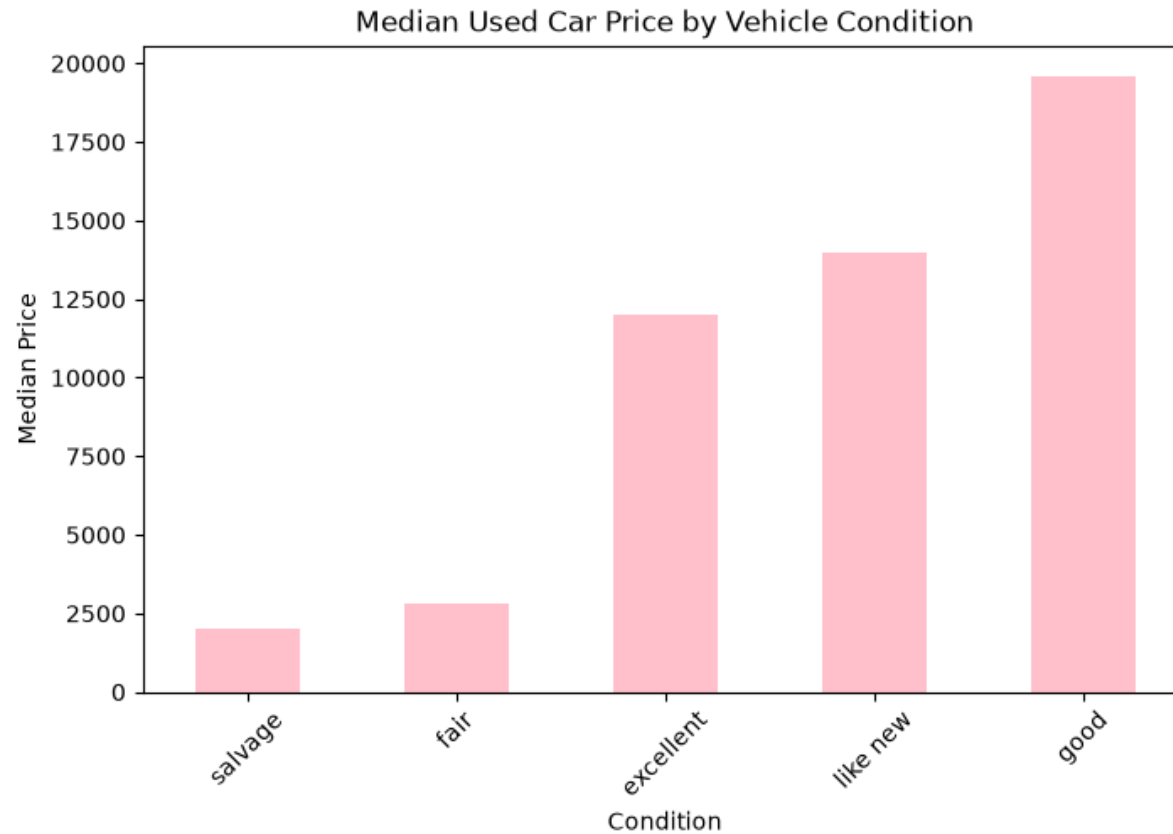
HOW DOES ODOMETER MILEAGE AFFECT VEHICLE PRICES?



Key Finding :

Odometer mileage has a strong effect on price. Cars with lower mileage generally have higher prices.





Key Finding :

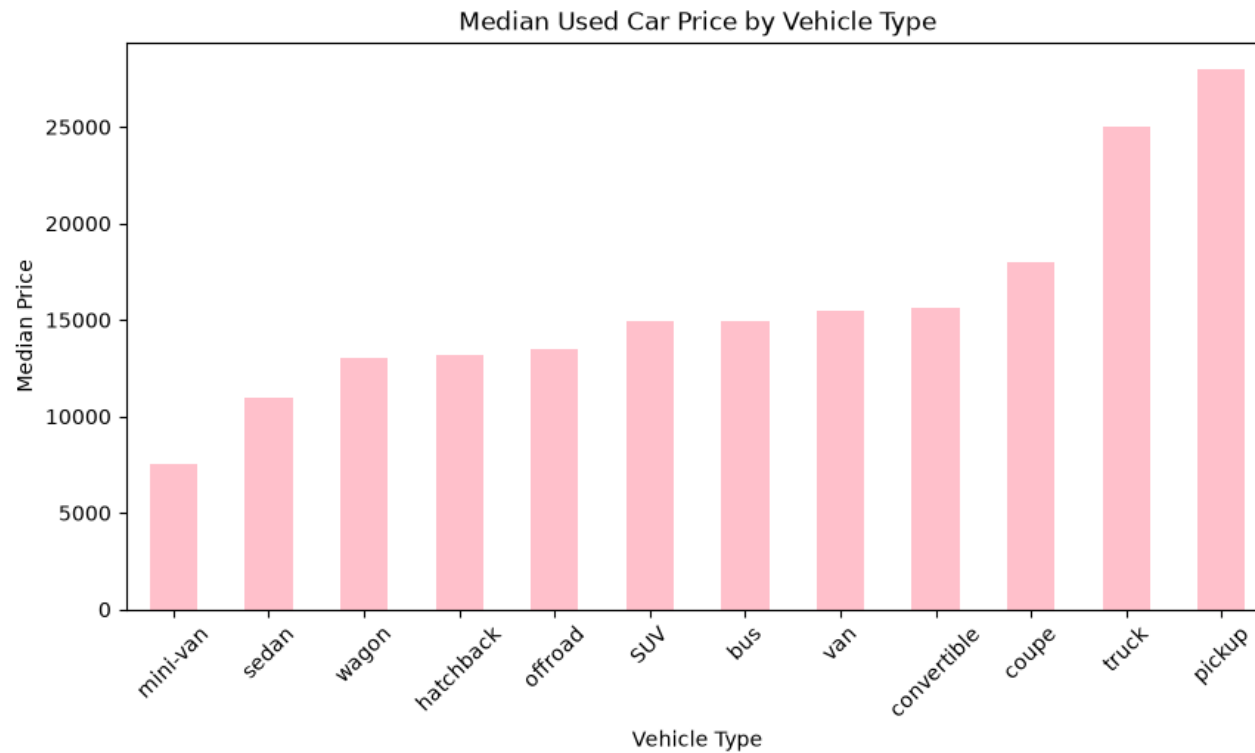
Used car prices vary depending on the vehicle condition, and cars in better condition generally have higher prices.

OBJECTIVE 4
HOW DO USED
CAR PRICES
VARY ACROSS
DIFFERENT
VEHICLE
CONDITIONS?



OBJECTIVE 5

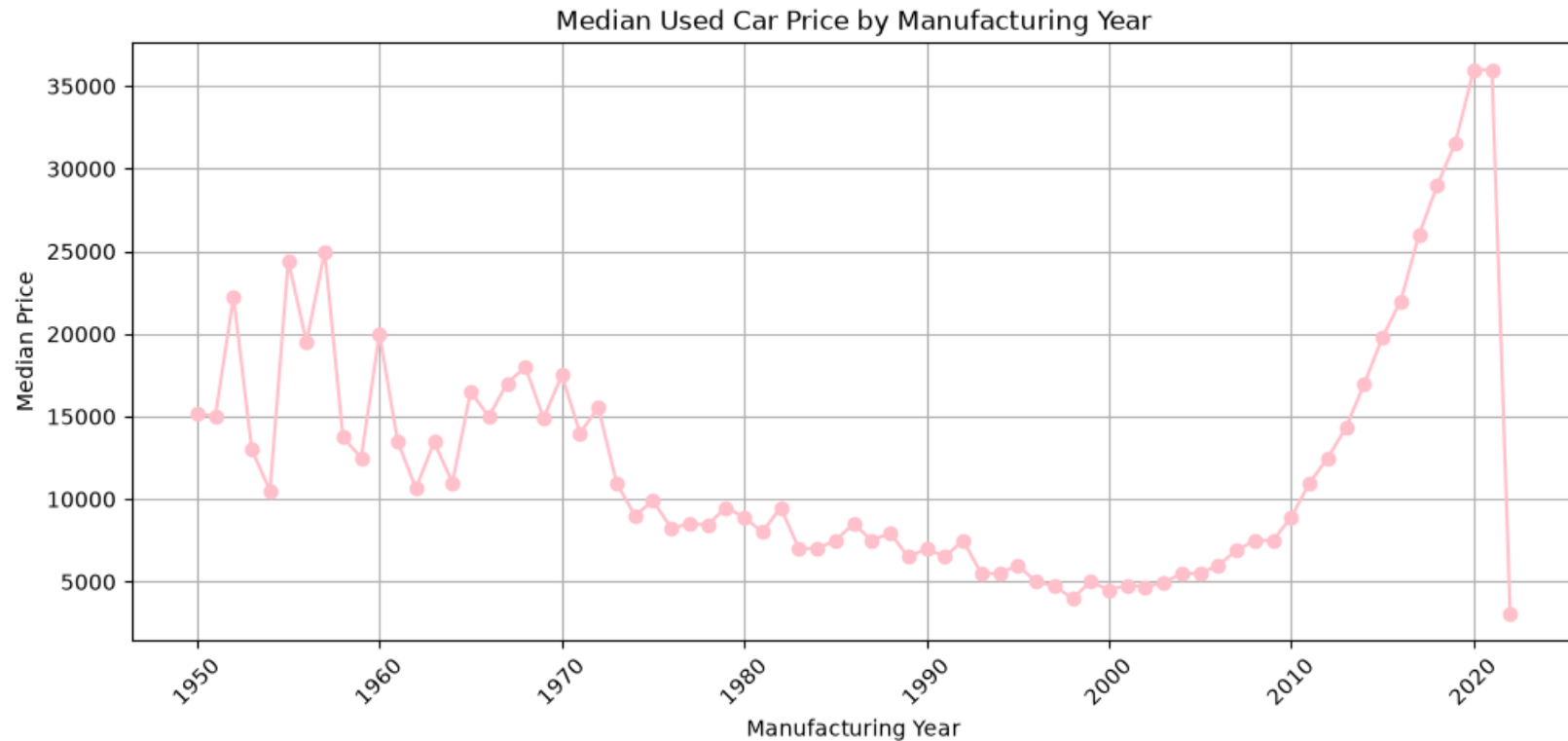
HOW DO USED CAR PRICES VARY ACROSS DIFFERENT VEHICLE TYPES?



Key Finding :

Vehicle type also has a noticeable effect on used car prices.



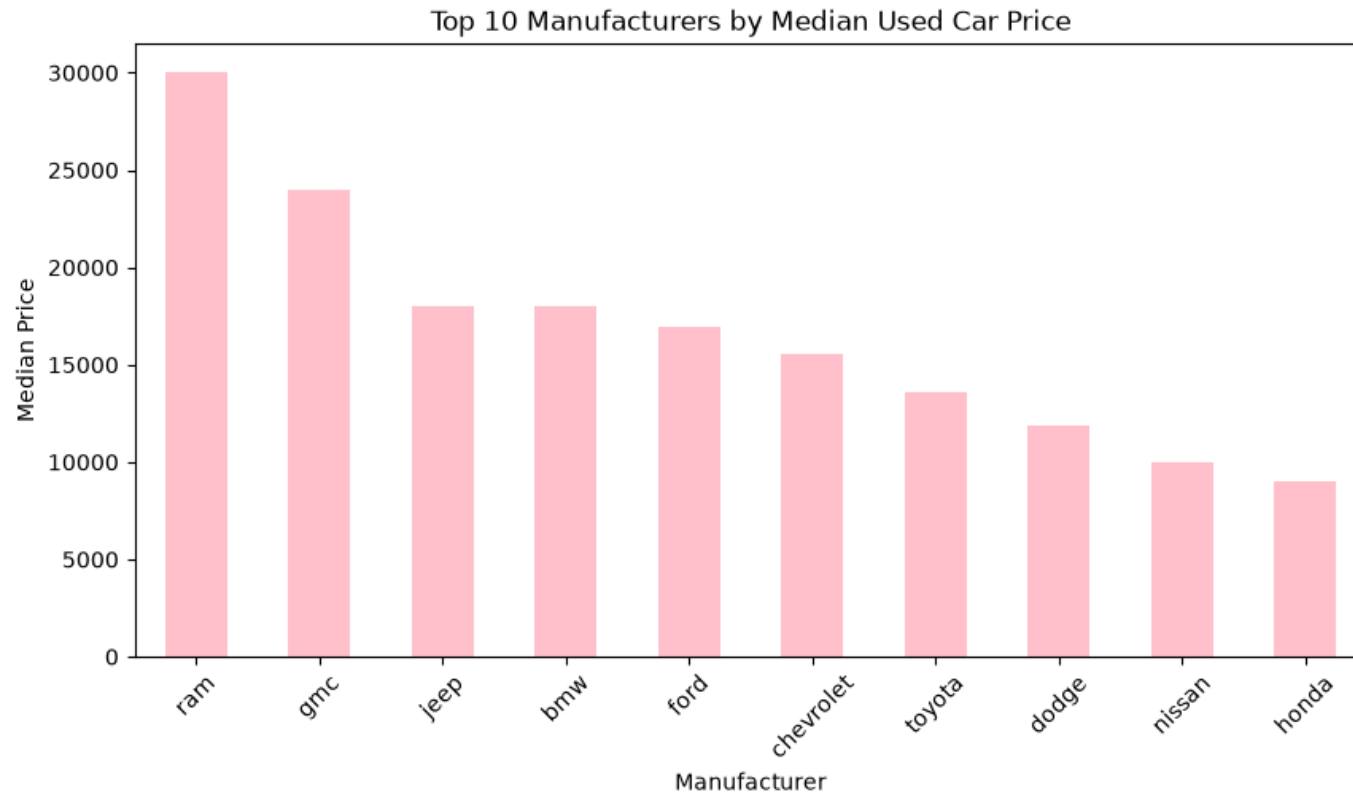


Key Finding :

Prices show a slight increase for very old vehicles, remain lower for some middle-year models, then increase significantly for newer vehicles. However, prices decrease slightly for the 2022 model year.

OBJECTIVE 6
HOW DOES THE
MANUFACTURING
YEAR AFFECT
VEHICLE PRICES?





OBJECTIVE 7
HOW DO USED CAR
PRICES VARY
ACROSS
DIFFERENT
MANUFACTURERS?



Key Finding :

The manufacturer has a strong effect on used car prices.

RECOMMENDATIONS :



Do not rely only on price when choosing a used car.



Compare similar cars before making a purchase.



Consider the manufacturing year and mileage when comparing vehicles.



Check the vehicle condition carefully.



Consider the vehicle's mileage, as cars with lower mileage generally have higher value and price.



Compare prices across different manufacturers to find the best value for your budget.





THANK YOU